

Decision Tree Algorithm Explained

A **Decision Tree** is a supervised machine learning algorithm used for **classification** and **regression** tasks. It works like a flowchart, where decisions are made by splitting the data into smaller subsets based on certain conditions.

1. How Decision Trees Work

A Decision Tree consists of the following components:

- **Root Node** → The starting point where the first decision is made.
- **Internal Nodes** → Each node represents a test on a feature.
- **Branches** → Represent possible outcomes of the test.
- **Leaf Nodes** → The final decision/output.

It follows these steps:

1. **Select the Best Feature** → The dataset is split using a feature that provides the most significant information gain (IG).
 2. **Split the Data** → The dataset is divided into subsets based on the feature's values.
 3. **Repeat Until a Stopping Condition is Met** → The tree grows until it reaches leaf nodes.
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2. How to Choose the Best Split?

To find the best feature to split on, we use **impurity measures** like:

A. Gini Impurity

Measures how "pure" a node is.

$$G = \sum_{i=1}^C p(i) * (1 - p(i))$$

Where $p(i)$ is the probability of a class appearing in a node. The lower the Gini, the purer the node.

B. Entropy (Information Gain)

$$H = - \sum_i p_i (\log_2 p_i)$$

A lower entropy means a better split.

C. Mean Squared Error (For Regression)

Used when predicting continuous values.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \tilde{y}_i)^2$$

3. Advantages & Disadvantages

Pros:

- Simple to understand and interpret.
- Can handle both numerical and categorical data.
- Requires little data preprocessing.

Cons:

- Can easily overfit (solved using pruning).
- Can be biased toward features with more levels.
- Sensitive to noisy data.